

Phil Donovan

Data Architecture · Engineering · Science · AI

DIRECTOR – VIRTUS SOLUTIONS

PRODUCTS – EOLAS.NZ

I design data platforms — Snowflake, dbt, Iceberg, AWS — and the models that run on them, for councils, health agencies and insurers. I came to it from geography, after ten years of spatial analysis across public health, transport and urban policy.

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Experience

2013 – PRESENT

2025 – now

Principal Consultant — Data Architecture, Engineering & AI

FIRN · AUCKLAND

- ▲ Led the design of Auckland Council's Modern Data Platform, centralising data assets for scale and governance.
- ▲ Built dbt, Snowflake and Fivetran pipelines — including automated ingestion of LINZ land parcels via WFS changesets, keeping council-wide spatial analysis current.
- ▲ Introduced and rolled out AI-assisted development tooling across the engineering team, setting the standards and workflows for its use.

2025 – now

Associate

PRINCIPAL ECONOMICS · AUCKLAND

- ▲ Built the Health Vulnerability and Adaptation dashboard for the Ministry of Health — an R Shiny and Plumber application modelling climate-change health impacts and adaptation options across New Zealand.
- ▲ Applied Treasury CBAX cost-benefit methodology, including declining discount rates, to cost adaptation scenarios down to SA2 level.
- ▲ Engineering on the Cost Intelligence Platform.

2024 – 2025

R/SQL Developer, Engineer & Scientist

AA INSURANCE · AUCKLAND

- ▲ Developed the pricing emulator that carried the migration from the legacy Protect platform to Duck Creek.
- ▲ Advised on the architecture of their AWS analysis platform, and built custom SageMaker container images so analysts were productive the moment one deployed.
- ▲ Wrote `aaair`, an R package joining Redshift and other sources to the pricing emulator for reproducible pricing simulation.

2023 – 2024	Senior Analyst, R Developer & SQL Engineer TE WHATU ORA – HEALTH NEW ZEALAND ▲ Led geospatial advisory for the Queenstown <i>Cryptosporidium</i> outbreak response, mapping contamination risk to direct public health intervention. ▲ Built <code>nphsintel</code>, an R package over Snowflake and PostGIS for population health analysis — cut code redundancy by around 75% and trained 10+ analysts on it. ▲ Introduced Quarto-based automated reporting and Shiny dashboards for spatial health trends.
2022 – 2023	Principal GIS Analyst MINISTRY OF HEALTH · WELLINGTON ▲ Automated geospatial workflows with the <code>mohanalytics</code> and <code>mohgeospatial</code> R packages, lifting team productivity by around 40%. ▲ Built and maintained the ArcGIS Online Portal in Python and R, surfacing interactive layers on health access disparities with ETL keeping them current. ▲ Wrote monitoring scripts tracking portal credit and cloud spend, cutting resource overruns by around 30%.
2018 – 2020	Spatial Data Scientist GEOGRAPHIC BUSINESS SOLUTIONS · AUCKLAND ▲ Deployed and configured the ESRI Enterprise stack — ArcGIS Pro and Server, Survey123, Collector, Operations Dashboard — for client geospatial platforms. ▲ Built 3D air pollution models for Seoul using the ArcGIS JavaScript API.
2016 – 2018	Spatial & Economic Data Analyst MRCAGNEY · AUCKLAND ▲ Delivered the National Policy Statement on Urban Development Capacity analysis — regression modelling of the effect of zoning regulation on land prices nationally. ▲ Ran the Whangārei residential and business development capacity assessment, building geometric algorithms to encode planning rules. ▲ Public transport modelling against the Auckland and Wellington transport models.
2013 – 2016	Spatial Data Analyst SHORE, MASSEY UNIVERSITY · AUCKLAND ▲ Principal spatial analyst on children's urban health research — from cleaning travel-diary data through to accessibility analysis. ▲ Co-authored five peer-reviewed papers on children's mobility and the built environment.

Capabilities	TOOLING
LANGUAGES	WAREHOUSES & ENGINES
R · Python · SQL	

Snowflake · Redshift · PostgreSQL · Oracle · SQL Server · DuckDB · Apache Arrow · Iceberg

PIPELINES

dbt · Meltano · Fivetran · Singer · PySpark

MODELLING

Regression · Supervised & unsupervised learning · Spatial models · Cost-benefit (CBAX)

APPS & REPORTING

R Shiny · Quarto · Power BI · ArcGIS Portal

CLOUD & INFRASTRUCTURE

AWS · Docker · Linux · Git

SPATIAL

PostGIS · ArcGIS · QGIS · FME · GRASS

Selected work

OPEN SOURCE & PROJECTS

BUILDING NOW

eolas

Official statistics for New Zealand, Australia and the OECD are scattered across dozens of portals, each with its own formats and quirks. eolas puts 700+ datasets behind a single typed API — clean columns, consistent schemas, no merged Excel headers.

I built the whole stack: Singer-spec taps and a Meltano orchestration layer landing data as Iceberg tables on S3 and Glue, exposed through a FastAPI service on AWS, with first-class Python and R clients, a CLI, and a Snowflake share for warehouse users.

PYTHON · R · SINGER · MELTANO · ICEBERG · FASTAPI · SNOWFLAKE · AWS

nzcensr

An R package that makes New Zealand census data straightforward to import, in both spatial and non-spatial form.

R · SPATIAL

NZ Economic Overview

An interactive Shiny dashboard over the vs-warehouse API, surfacing New Zealand economic indicators.

R · SHINY

NPS-UDC

National Policy Statement on Urban Development Capacity — quantifying how local government regulation moves land prices across New Zealand's major urban areas, by regression on zoning rules.

ECONOMICS · SPATIAL REGRESSION

NDAI-C

The Network Destination Accessibility Index for Children: a tool measuring how reachable everyday destinations are for kids across Auckland, Hamilton, Wellington and Christchurch.

ACCESSIBILITY · NETWORK ANALYSIS

nphsintel

An R package for Te Whatu Ora giving analysts one route into population health data across Oracle and Snowflake, with spatial analysis built in. Cut code redundancy by around 75%.

R · SNOWFLAKE · POSTGIS

Writing

2023 — 2026

21 July 2026

Building eolas: a data platform for New Zealand official statistics

Why official data is tedious to work with, how the platform is put together, and which decisions turned out to matter.

24 December 2025

The quiet power of S3

Vaccine-specific summaries from one event table, using R's smallest object system.

24 December 2025

Serving R charts with Plumber and HTMX

A Highcharts figure drawn in R, returned as HTML, and swapped into the page.

16 May 2024

Enhancing efficiency with the `nphsintel` R package

Why an in-house R package was worth building, and how it cut analysis code by more than 75%.

7 January 2024

Utilising S3 classes in R for vaccine data analysis

Using R's oldest object system to make messy immunisation data behave.

29 November 2023

Packaging your day-to-day data analysis

Writing an R package is not reserved for virtuosos — it's the cheapest way to stop repeating yourself.

16 July 2023

Demystifying Apache Arrow with R

What Arrow actually is, and why it matters once your data outgrows memory.

Education

2007 — 2013

2012 — 2013

MSc in Geography — *First Class Honours*

University of Auckland

2011

PgDip in Geography

University of Auckland

2007 — 2010

BSc in Geography and Geology

Victoria University of Wellington

01

Kids in the City: Children's Use and Experiences of Urban Neighbourhoods in Auckland, New Zealand
Carroll, Witten, Kearns & Donovan

02

Associations between the neighbourhood built environment and out of school physical activity and active travel
Oliver, Mavoa, Badland, Parker, Donovan, Kearns, Lin & Witten

03

Children's Out-of-School Independently Mobile Trips, Active Travel, and Physical Activity
Oliver, Parker, Witten, Mavoa, Badland, Donovan, Chaudhury & Kearns

04

Assessing neighbourhood destination access for children: development of the NDAI-C audit tool
Badland, Donovan, Mavoa, Oliver, Chaudhury & Witten

05

Development of a systems model to visualise the complexity of children's independent mobility
Badland, Kearns, Carroll, Oliver, Mavoa, Donovan, Parker, Chaudhury, Lin & Witten

Get in touch

Open to conversations about data platform work, spatial architecture, and anything where the two meet.

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